



ARTIFICIAL INTELLIGENCE · LECTURE 03

Solving Problems the AI Way

How to turn a real problem into something an AI can actually solve

Learning Objectives



Understand why a computer needs a problem “translated” before it can solve it.



Learn the 5 parts every well-defined problem has.



Formulate a real-life problem (like a road trip) as a search problem.



Understand what a “state space” is, using a simple diagram.



Compare toy problems (like the 8-Puzzle) with real-world problems.



Understand abstraction — deciding what detail to keep or leave out.



Understand path cost and how AI picks the “best” solution.

Why Can't We Just Say “Figure It Out”?

A computer can't solve a problem it doesn't understand — first we must describe it precisely.

Imagine telling a friend: “Get from Lahore to Islamabad.” A human just... does it. A computer needs the problem broken into exact pieces first — that process is called Problem Formulation.

A Problem-Solving Agent Follows 4 Simple Steps:

- 1 Goal Formulation — decide what you actually want (“be in Islamabad”)
- 2 Problem Formulation — decide what states, actions & costs matter
- 3 Search — explore possible action sequences to find a solution
- 4 Execution — carry out the chosen sequence of actions

5 Parts of Any Well-Defined Problem

Just like PEAS described an agent, these 5 questions describe any problem.



Initial State

Where does the agent start?



Actions

What moves/choices are available at each point?



Transition Model

If I take this action, what state do I end up in?



Goal Test

How do I know I've reached the goal?



Path Cost

How “expensive” (time, money, distance) is a solution?

WORKED EXAMPLE

Route-Finding: Lahore → Islamabad

Let's formulate a road trip as a search problem



Initial State

“In Lahore”



Actions

Drive to any directly connected city (e.g., Gujranwala, Sialkot)



Transition Model

Taking a road from City A leads to City B



Goal Test

“Is the current city Islamabad?”



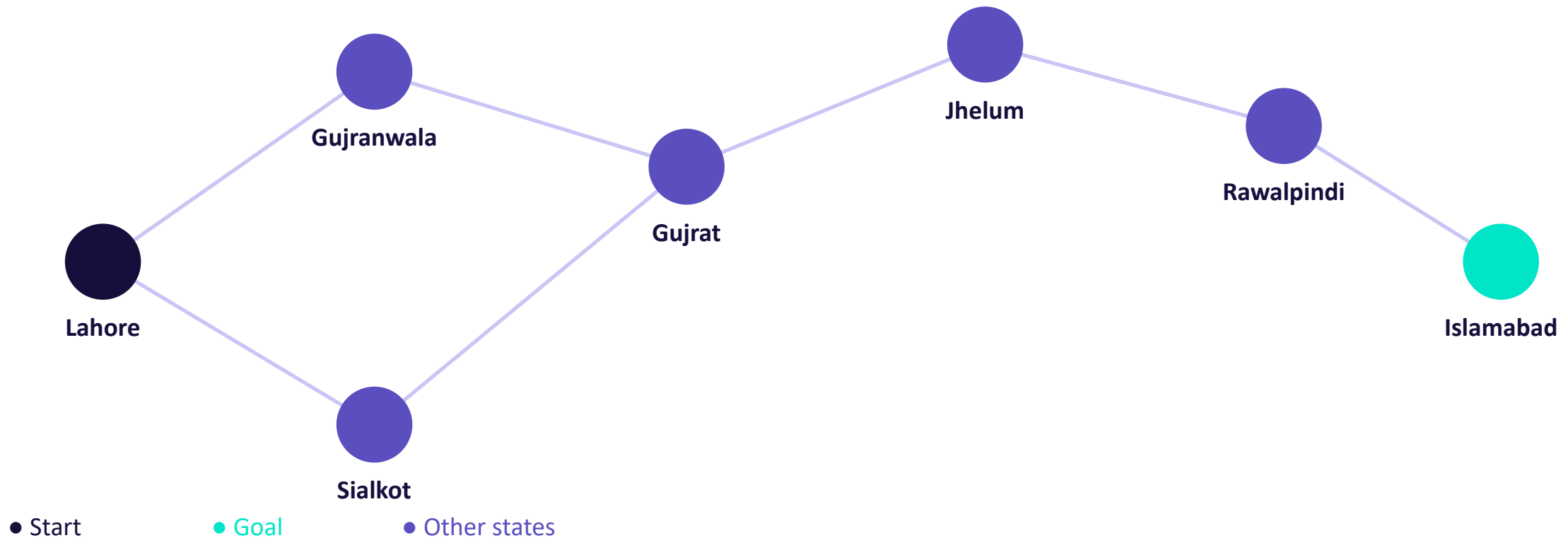
Path Cost

Total distance (km) or total travel time

VISUALIZING IT

The State Space — All Possible Situations

Every city becomes a dot (node); every road becomes a line (edge).



A path from Lahore to Islamabad through this graph is one possible solution.

Toy Problems vs. Real-World Problems

Two categories researchers use to study and test problem-solving methods.



Toy Problems

Simple, precisely defined problems used to test and compare algorithms fairly.

- 8-Puzzle — slide numbered tiles into order
- 8-Queens — place queens so none attack each other
- Vacuum World — our earlier example
- Missionaries & Cannibals — classic river-crossing puzzle



Real-World Problems

Messier, harder to define precisely, but this is where AI creates real value.

- Route-Finding — GPS apps like Google Maps
- Touring / Delivery Problems — visit many cities once (TSP)
- Robot Navigation — moving through real, messy spaces
- Assembly Sequencing — order of steps in a factory

■ SECOND EXAMPLE

The Vacuum-Cleaner World

AIMA's classic simple example — two rooms, one cleaning robot.



Initial State

In Room A, and both rooms are Dirty



Actions

Move Left, Move Right, Suck (clean current room)



Transition Model

“Suck” changes Dirty → Clean in current room



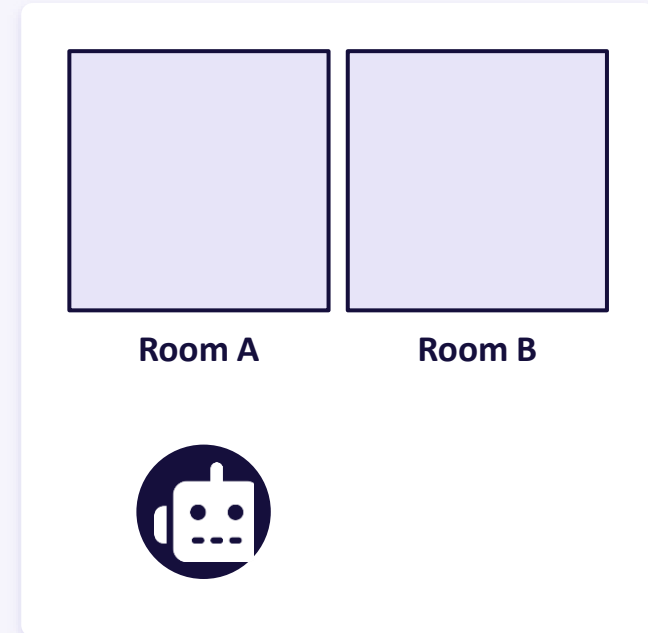
Goal Test

Are both Room A and Room B Clean?



Path Cost

Number of actions taken (fewer is better)



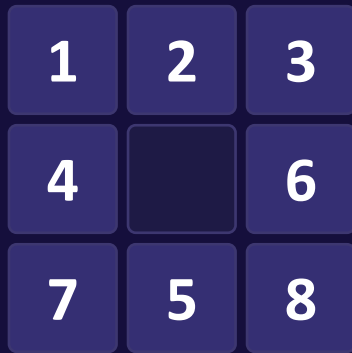
The robot can be in either room, and each room can be Clean or Dirty — giving 8 possible states in total.

THIRD EXAMPLE

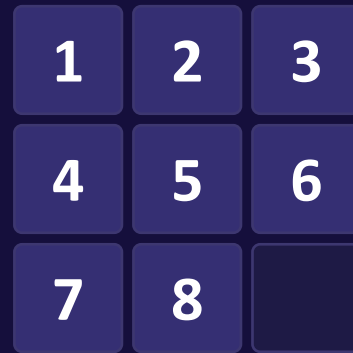
The 8-Puzzle

A classic toy problem — slide tiles to match the goal picture

START STATE



GOAL STATE



Initial State

Any starting arrangement of the 8 numbered tiles



Actions

Slide the blank tile Up, Down, Left, or Right



Transition Model

Moving blank swaps it with the adjacent tile



Goal Test

Do the tiles match the goal arrangement (1–8 in order)?



Path Cost

Number of tile moves made (each move = cost 1)



Even this simple puzzle has 181,440 reachable states — showing why AI needs smart search, not guesswork.

Abstraction — The Art of Leaving Things Out

A good formulation keeps what matters and throws away the rest.



Abstraction means removing irrelevant detail from a problem while keeping it accurate enough to solve.

Example: Lahore → Islamabad, Revisited

KEPT (relevant to the goal)

- ✓ Which cities connect to which
- ✓ Distance or travel time of each road

LEFT OUT (irrelevant to the goal)

- ✗ Color of the cars on the road
- ✗ Scenery, weather, radio stations
- ✗ Names of petrol pumps passed

Too much detail → the problem becomes too slow to solve. Too little → the solution may not work in reality. Good abstraction finds the balance.

Path Cost — How AI Judges “Best”

Not every solution is equally good — path cost tells us which one wins.

A path cost function adds up a small “step cost” for every action taken along a path.

Lahore → Gujranwala → Gujrat → Jhelum → Rawalpindi → Islamabad

Step costs: $70 + 60 + 90 + 100 + 30 = 350$ km total — this route's path cost.

Different routes can reach the SAME goal with DIFFERENT costs — a rational agent prefers the lowest-cost (or otherwise best) solution.

Cost doesn't have to be distance — it could be time, money, battery used, or even risk.

■ AVOID THESE

Common Mistakes in Problem Formulation

- Too much detail — including irrelevant facts (like the color of the car) makes the problem needlessly complex.
- Too little detail — leaving out something the goal test actually needs (e.g., forgetting fuel level in a low-fuel scenario).



● Vague goal test — “get there eventually” isn't testable; “current city = Islamabad” is.

● Forgetting path cost — without it, the agent can't tell a great solution from a terrible one.



TRY IT YOURSELF



Formulate Your Own Problem

- 1 In pairs, pick ONE: getting from your home to campus, solving a maze, or packing a suitcase.
- 2 Write its Initial State, Actions, Transition Model, Goal Test, and Path Cost.
- 3 Ask ChatGPT to check your formulation and suggest anything you missed.
- 4 Be ready to explain your Goal Test and Path Cost to the class.

■ WRAPPING UP

Key Takeaways

- Before an AI can solve a problem, it must be formulated precisely — humans can't just say “figure it out.”
- Every well-defined problem has 5 parts: Initial State, Actions, Transition Model, Goal Test, and Path Cost.
- Toy problems (8-Puzzle, Vacuum World) test algorithms fairly; real-world problems (route-finding, TSP) are where AI creates value.
- Abstraction means keeping only the details relevant to the goal — too much or too little detail both cause problems.
- The state space is the map of every possible situation the agent could be in, connected by actions.
- Path cost lets a rational agent compare solutions and pick the best one, not just any one.



Recommended Reading



Primary Text

Russell, S. & Norvig, P. (2010). Artificial Intelligence: A Modern Approach (3rd ed.). Chapter 3 (Solving Problems by Searching).



Reference

Luger, G.F. & Stubblefield, W.A. (2009). AI Algorithms, Data Structures, and Idioms in Prolog, Lisp, and Java — chapters on search.

NEXT LECTURE

Uninformed Search: BFS, DFS & Uniform-Cost Search